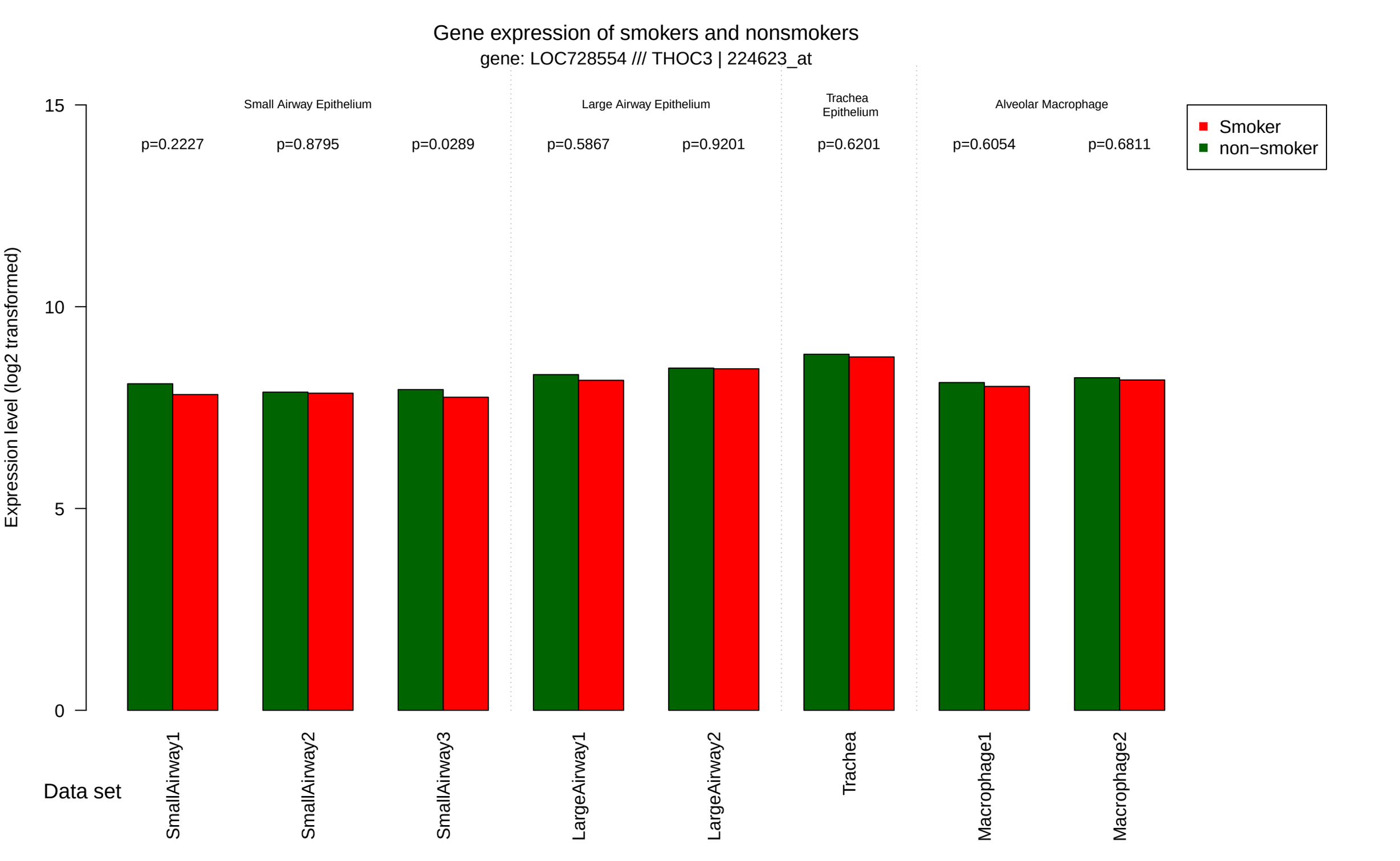
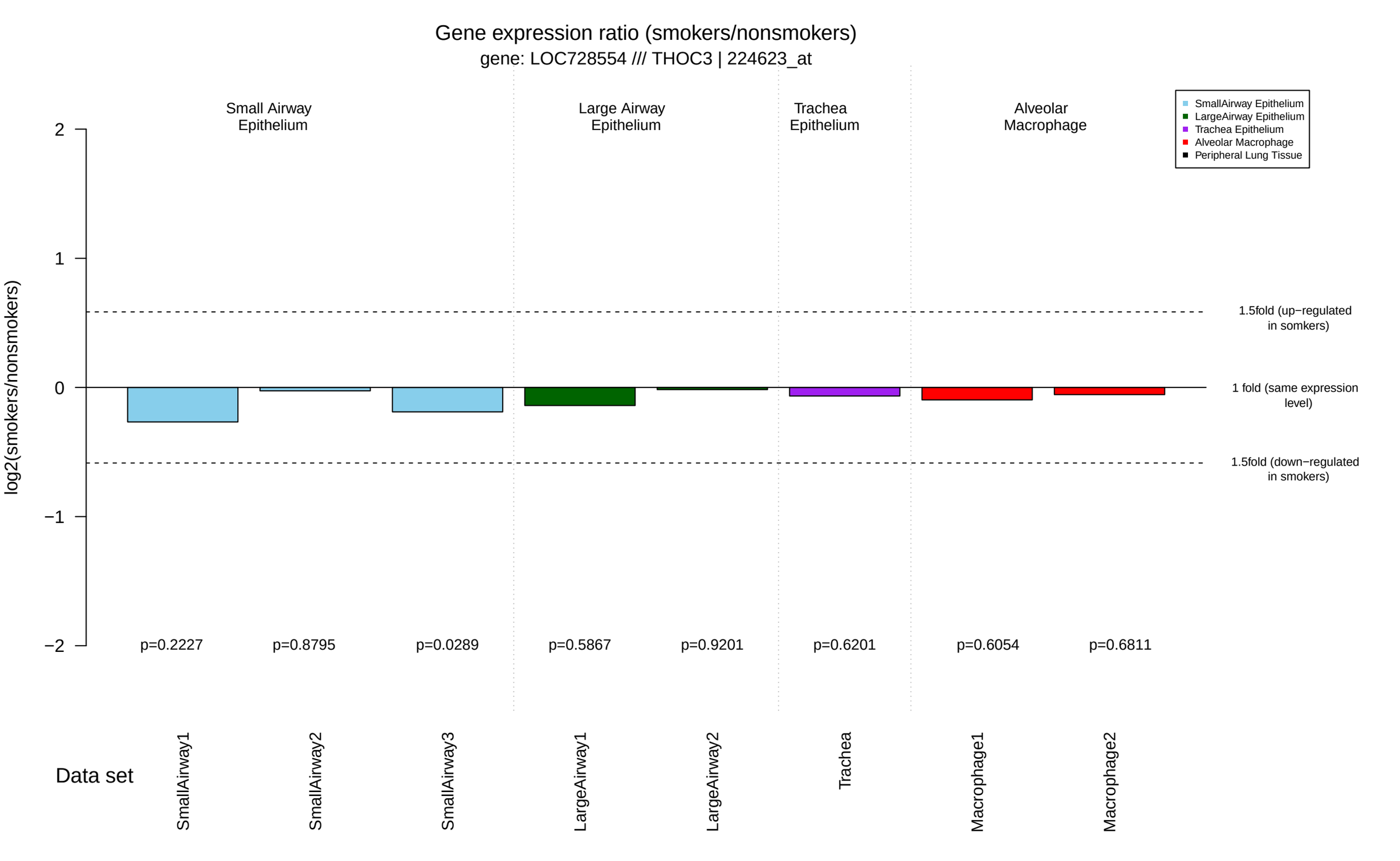
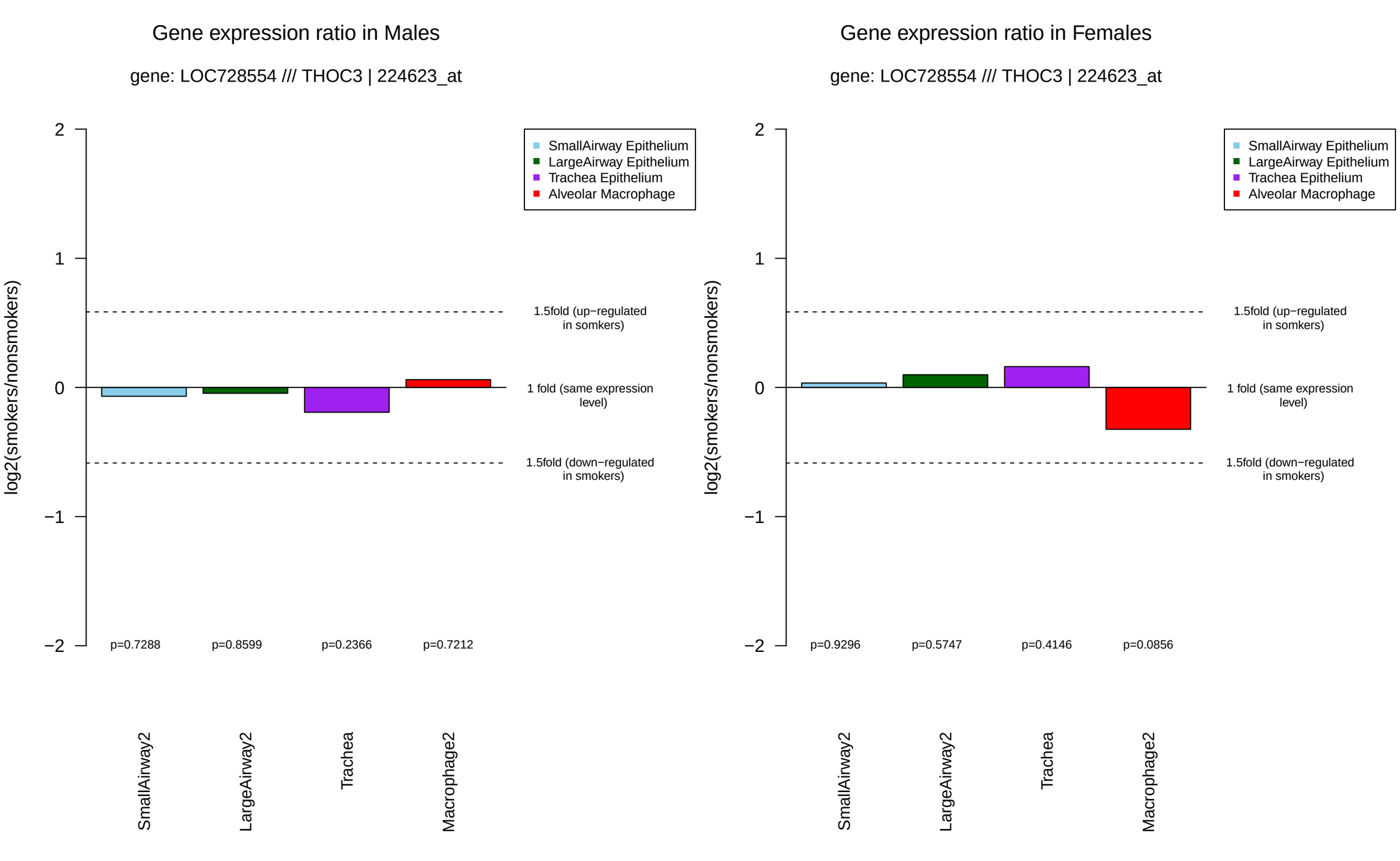
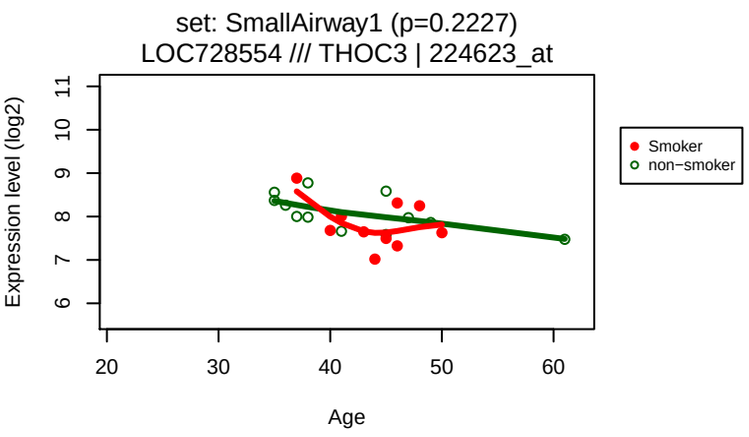
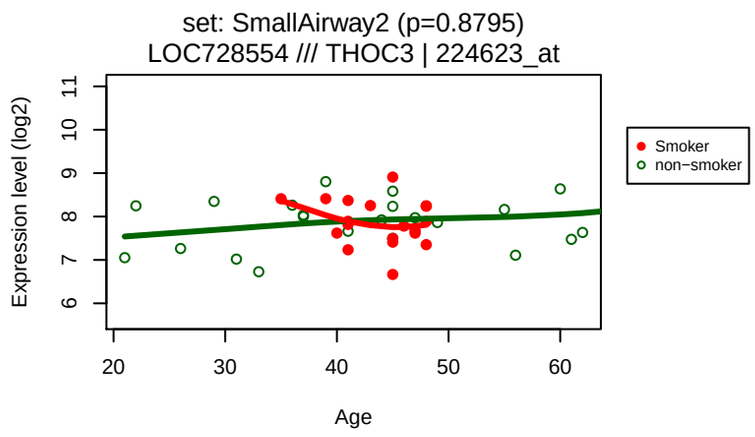
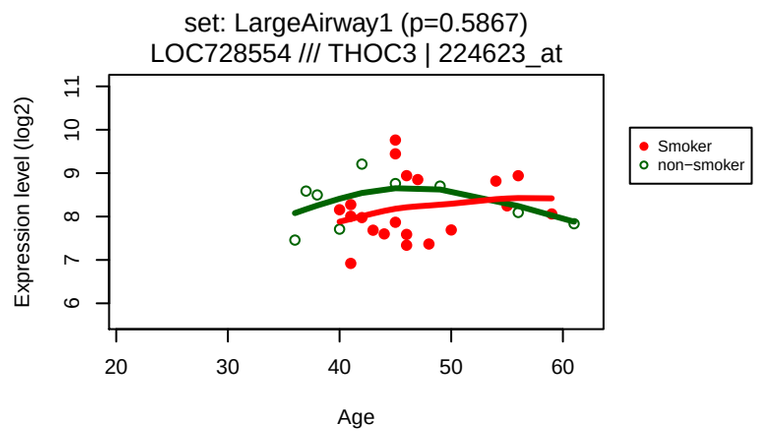

Gene expression of smokers and nonsmokers
gene: LOC728554 /// THOC3 | 224623_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2227
p=0.8795
p=0.0289
p=0.5867
p=0.9201
p=0.6201
p=0.6054
p=0.6811
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

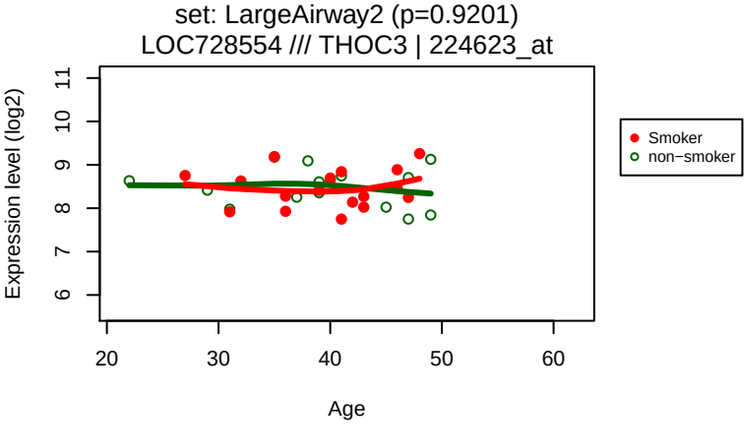

Gene expression ratio (smokers/nonsmokers)
gene: LOC728554 /// THOC3 | 224623_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2227
p=0.8795
p=0.0289
p=0.5867
p=0.9201
p=0.6201
p=0.6054
p=0.6811
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

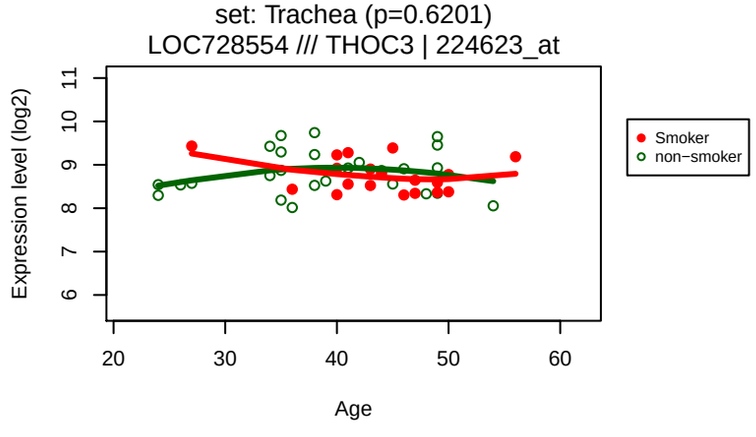

Gene expression ratio in Males
gene: LOC728554 /// THOC3 | 224623_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7288
p=0.8599
p=0.2366
p=0.7212
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: LOC728554 /// THOC3 | 224623_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9296
p=0.5747
p=0.4146
p=0.0856
SmallAirway2
LargeAirway2
Trachea
Macrophage2

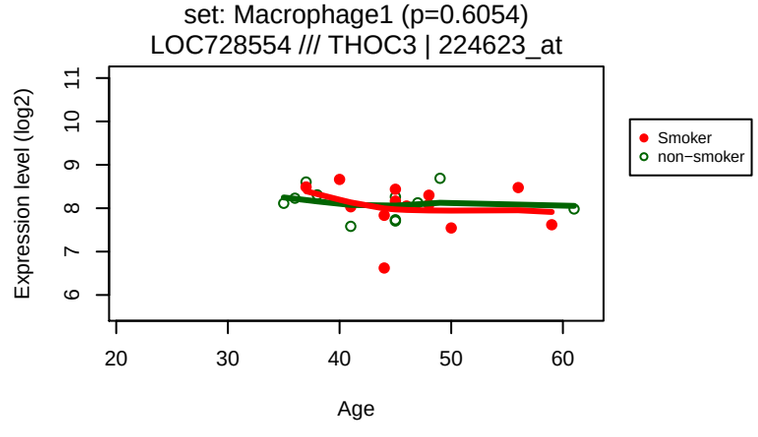

set: SmallAirway1 (p=0.2227)
LOC728554 /// THOC3 | 224623_at
Expression level (log2)
Age
Smoker
non−smoker

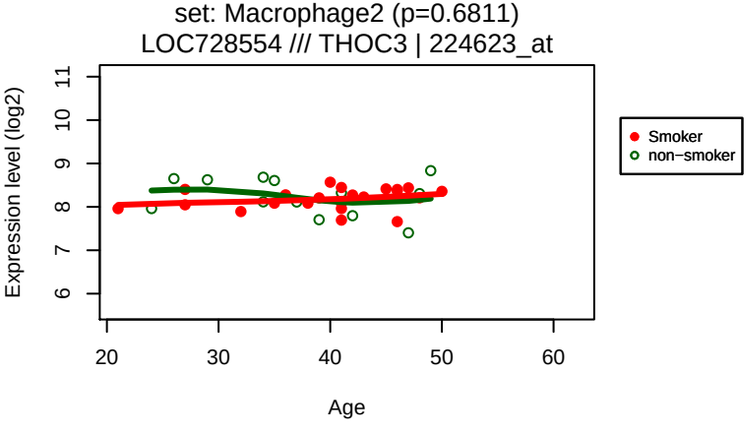

set: SmallAirway2 (p=0.8795)
LOC728554 /// THOC3 | 224623_at
Expression level (log2)
Age
Smoker
non−smoker

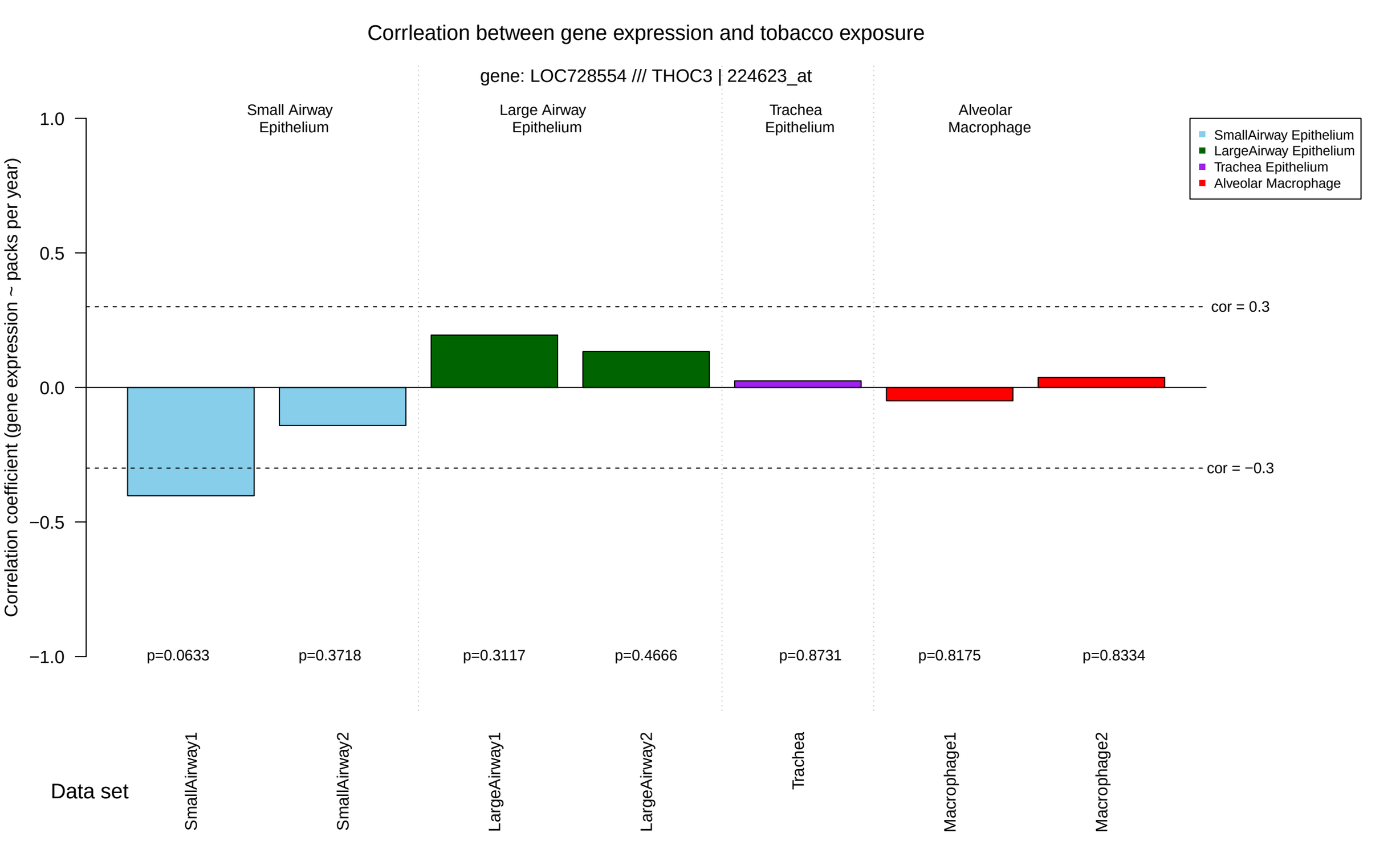

set: LargeAirway1 (p=0.5867)
LOC728554 /// THOC3 | 224623_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.9201)
LOC728554 /// THOC3 | 224623_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.6201)
LOC728554 /// THOC3 | 224623_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.6054)
LOC728554 /// THOC3 | 224623_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.6811)
LOC728554 /// THOC3 | 224623_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: LOC728554 /// THOC3 | 224623_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0633
p=0.3718
p=0.3117
p=0.4666
p=0.8731
p=0.8175
p=0.8334
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2